Zagazig University
Faculty of Computers & Informatics
Final Term Exam Academic Year: 2023-2024

Department of Computer Science
Study Year/Level: First Year
Course Title: Discrete Structures
Course Code: CS103

Examination Date: 01/06/2024
Examination Starts: Morning
Allowed Examination Time: 3 Hours
Total Marks: 90 Marks

Model 1

Student Name: **Student ID:**

- **Authorized Materials:**
- **Calculators allowed. No other additional materials or aides allowed.**
- **Instructions to Invigilators:**
- **The exam paper may be taken out of the examination room, but after half time.**
- **Instructions to Students:**
- **If you feel ill, let us know immediately.**
- **Make sure your name is on every page.**
- **Answer All of the following questions.**
- **If you get stuck on a question, move on and come back later if you have time.**
- **Each equation has only one correct answer.**
- **Write your solutions directly into the exam booklet.**
- **You can obtain 90 points. You need 27 points to pass.**
- **Remember to time yourself.**

Enjoy!

MULTIPLE CHOICE. Choose the one alternative that BEST completes the statement or answers the question.

1. Is $\{4\} \in \{2, 4, 8\}$? a. True b. False

2. Is $\{4\} \subseteq \{\{2\}, \{4\}, \{8\}\}$? a. True b. False

3. Is $\phi \subseteq \{\phi\}$? a. True b. False

4. Is $A \subseteq B \leftrightarrow \forall x\ (x \in A \rightarrow x \in B)$? a. True b. False

5. Let A, and B two sets, $A \cap (A \cup B) = A$ a. True b. False

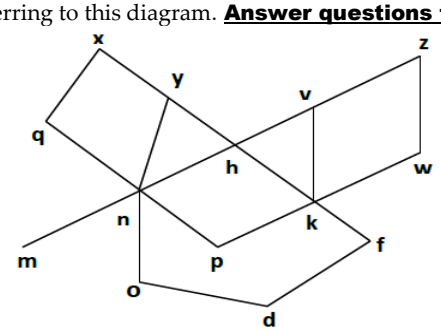
6. Let A = {x, y, z} and B = {u, v}. Then, B × A = {(x, u); (x, v); (y, u); (y, v); (z, u); (z, v)} a. True b. False

7. Let A = {3, 5}, B = {x, y, z}, and C = {3, 4, 5}. Find the cardinality of (A × B × C):
a. 9 b. 8 c. 16 d. 18 e. None

8. Find the cardinality of $\{x \mid x \text{ is an even integer and } 4 \le x \le 20\}$:
a. 7 b. 8 c. 9 d. 4 e. None

9. Find the cardinality of P({1, 2}):
a. 7 b. 8 c. 9 d. 4 e. None

The diagram shown is the Hasse diagram for a partially ordered set. Referring to this diagram. **Answer questions from 10-19**



10. The maximal elements are:
a. q,x,z b. x,y,v,z c. y,v d. x,z e. None

11. The minimal elements are:
a. d,m,p b. o,d,f,m c. d,m d. x,z e. None

12. All upper bounds for w,n are:
a. k b. h c. z d. x e. None

13. All upper bounds for f,q are:
a. k b. h c. z d. x e. None

14. All lower bounds for y,w are:
a. k,p,f,d b. k,p,f c. d,f d. f,p e. None

15. The lub({n,p,d}) is:
a. d b. k c. n d. f e. None

16. The glb({y,v}) is:
a. x b. z c. h d. k e. None

17. The greatest element is:
a. x b. z c. y d. v e. None

18. The least element is:
a. m b. d c. o d. f e. None

19. One of the topological sorting for the poset is: d,o,f,p,m,k,n,m,w,v,q,z,x a. True b. False

Determine if the sequence defined by the formula is explicit or recursive. In addition, identify the value of the 3rd element in the sequence. **Answer questions from 20-21**

20. $a_n = 2n-1$ a. explicit, $a_3 = 5$ b. recursive, $a_3 = 5$ c. explicit, $a_3 = 7$ d. recursive, $a_3 = 7$

21. $C_1 = 2;\ C_n = 2C_{n-1}$ a. explicit, $C_3 = 8$ b. recursive, $C_3 = 8$ c. explicit, $C_3 = 4$ d. recursive, $C_3 = 4$

Suppose $A=\begin{bmatrix}0&0&1\\1&1&0\\1&0&0\end{bmatrix}$, $B=\begin{bmatrix}0&1&1\\1&1&0\\1&0&1\end{bmatrix}$ and $A \odot B = \begin{bmatrix}*&*&**&*&**&*&*\end{bmatrix}$.

Answer questions from 22-26

22. The first row of the matrix of $A \odot B$ is 101 a. True b. False

23. The last row of the matrix of $A \odot B$ is 111 a. True b. False

24. The first row of the matrix of $A \vee B$ is 011 a. True b. False

25. The last row of the matrix of $A \vee B$ is 101 a. True b. False

26. The first row of the matrix of $A \wedge B$ is 101 a. True b. False

27. If A= {2, 6, 8, 9} and B= {1, 2, 6, 8}, $A \oplus B$= {1, 9}
a. True b. False

28. If Let A = {a, b, c, d, e, f, g, h} and A2, A3, A4, A5 are subsets of A. A2 = {a, c, e, f, g, h}, A3 = {a, c, e, g}, A4 = {b, d}, A5 = {f, h}. Is {A3, A4, A5} a partition of A? a. True b. False

29. For any two finite non-empty sets A and B, $|A \times B| = |A| \times |B|$ a. True b. False

The relation *R* on {1, 2, 3, …} where *aRb* means a|b. **Answer questions from 30-33**

30. Is R reflexive? a. True b. False

31. Is R symmetric? a. True b. False

32. Is R antisymmetric? a. True b. False

33. Is R transitive? a. True b. False

The relation R on Z where aRb means $|a - b| \le 1$. **Answer questions from 34-35**

34. Is R Irreflexive? a. True b. False

35. Is R not transitive? a. True b. False

Suppose R and S are relations on {a,b,c,d}, where R = {(a,b),(a,d),(b,c),(c,c),(d,a)} and S = {(a,c),(b,d),(d,a)}. **Answer the following TRUE/FALSE questions from 36-53**

36. R^2 = {(a,a),(a,c),(b,c),(c,c),(d,b),(d,d)}.

37. R^{-1} = {(b,a),(d,a),(c,b),(c,c),(d,a)}.

38. R^3 = {(a,b),(a,c),(a,d),(b,c),(c,c),(a,d),(d,c)}.

39. S^2 = {(b,a),(d,c),(b,d)}.

40. S^5 = {(b,c)}.

41. Dom (R) = {a,b,c,d}.

42. Ran (S) = {a,b,d}.

43. In R, in-degree for the vertex c = 3.

44. In S, out-degree for the vertex c = 1.

45. R(a) = {b,d}.

46. $(S^{-1})^{-1}$ = {(a,c),(b,d),(d,a)}.

47. Ran (R) = Dom (R^{-1}).

48. Is S an identity relation

49. Is R a universal relation

50. Is S a void relation

51. R^{∞} = {(a,b),(a,c),(a,d),(b,c),(c,c)}.

52. R^* = {(a,b),(a,c),(a,d),(b,c),(c,c),(a,a),(b,b),(d,d)}.

53. $S \circ R$ = {(a,c),(b,a),(b,b),(d,b)}.

Consider the relation R on the set {a, b, c, d}. **Answer questions from 54-55**

54. What is the matrix that represents the asymmetric relation?

a. $\begin{bmatrix}1&0&1&1\\0&1&0&0\\1&0&1&1\\1&0&1&1\end{bmatrix}$ b. $\begin{bmatrix}0&1&1&0\\1&1&0&0\\0&0&0&1\\1&0&0&0\end{bmatrix}$ c. $\begin{bmatrix}0&1&1&0\\1&0&0&0\\0&0&0&1\\1&0&0&0\end{bmatrix}$ d. $\begin{bmatrix}0&0&1&0\\0&0&0&0\\0&0&0&1\\1&0&0&0\end{bmatrix}$

55. What is the matrix that represents the transitive relation?

a. $\begin{bmatrix}0&1&1&1\\0&0&1&1\\1&0&0&1\\0&1&0&1\end{bmatrix}$ b. $\begin{bmatrix}0&1&1&0\\1&1&0&0\\0&0&0&1\\1&0&0&0\end{bmatrix}$ c. $\begin{bmatrix}1&1&1&1\\0&0&1&1\\0&0&1&1\\0&0&0&1\end{bmatrix}$ d. $\begin{bmatrix}1&1&1&1\\1&0&1&0\\1&1&1&1\\1&0&1&1\end{bmatrix}$

56. Consider the R on the set {a, b, c, d}= $\begin{bmatrix}1&0&0&0\\1&1&0&0\\0&0&1&0\\0&0&1&1\end{bmatrix}$,

Is R an Equivalence relation? a. True b. False

57. Let A={x,y,z,w} and let R = {(x,x), (x,y), (y,x), (y,y), (z,z), (z,w), (w,w), (w,z)}. Is the equivalence classes A/R = {{x,y}} a. True b. False

Let A = {1, 2, 3, 4} and let R = {(1,4), (2,1), (2,3), (3,1), (3,4), (4,3)} be a relation to A. **Answer questions from 58-59**

58. Which of the following elements is in the transitive closure of R? a. (3,2) b. (2,2) c. (1,2) d. (4,1) e. None

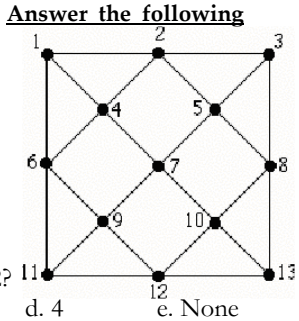
59. Which of the following elements is in the reflexive closure of R? a. (1,2) b. (1,3) c. (2,2) d. (4,2) e. None

60. Warshall algorithm is considered a more efficient method for computing the transitive closure of a relation? a. True b. False

61. An undirected graph G which is connected and acyclic is called_.
a.bipartite graph b.cyclic graph c. tree d. forest

62. A Partially ordered set is a set A with a relation R on it which is reflexive, anti-symmetric, and transitive? a. True b. False

Consider the graph G shown below. **Answer the following TRUE/FALSE questions from 63-70**



63. Does it have an Euler circuit?

64. Does it have an Euler path?

65. G is connected.

66. G is a simple graph.

67. Does it have an isolated vertex?

68. Does it have a pendant vertex?

69. What is the degree of the vertex 12?
a. 3 b. 2 c. 1 d. 4 e. None

70. Using Fleury's algorithm to find an Euler path in G, one possible solution is: 1,2,4,6,9,11,12,13,10,7,8,5,3.

71. A simple graph called regular if every vertex of the graph has the same degree. a. True b. False

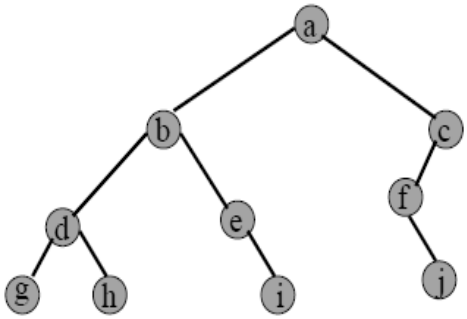
72. The complete graph has n vertices and n(n-1)/2 edges. a. True b. False

Let A = {1, 2, 3, 4, 5} and R = {(1, 2), (1, 3), (2, 4), (3, 5)}. **Answer the following questions from 73-74**

73. Is **R** a rooted tree a. True b. False

74. If R is a rooted tree, the root is: a.5 b.4 c.2 d.1 e.None

Consider the following tree (T) shown below. **Answer the following questions from 75-84**



75. What is the height of T? a. 2 b.1 c. 3 d. 4

76. T is an n-tree. What is the value of n? a. 1 b.5 c. 3 d. 2

Answer the following TRUE/FALSE questions from 77-81

77. Is (T) a balanced tree?

78. Is (T) a complete 2-tree?

79. Preorder Traversal of (T) is: a b d g h e i c f j

80. Inorder Traversal of (T) is: g d h b e i a f c j

81. Postorder Traversal of (T) is: g h d i e b j f c a

82. The number of leaves in (T) is: a. 3 b. 6 c. 4 d. 2

83. The siblings of d is: a. g b. h c. f d. e

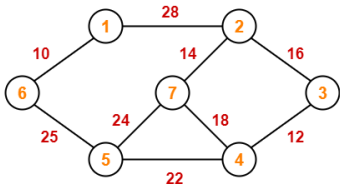
84. The descendants of c are: a. c,f,j b. f c. f,j d. a

Consider a complete 3-tree with 4 internal vertices. **Answer questions from 85-86**

85. What is the number of vertices? a.19 b.7 c.10 d. 13

86. Based on the previous answer for the number of vertices, what is the number of leaves? a.13 b.5 c. 9 d. 7

Using Prim algorithm, find the MST starting from vertex 6. **Answer questions from 87-90**



87. Which vertex will be selected at iteration 2? a.2 b.5 c.7

88. Which vertex will be selected at iteration 5? a.3 b.4 c.2

89. The cost of the minimum spanning tree is:
a. 96 b. 109 c. 103 d. 99 e. None

90. The number of edges that are not selected in MST is:
a. 2 b. 1 c. 3 d. 4 e. None

Good Luck *Dr. Abduallah Gamal*